OpticPS: An Open-Source Framework for Probabilistic Shaping in Optical Communications

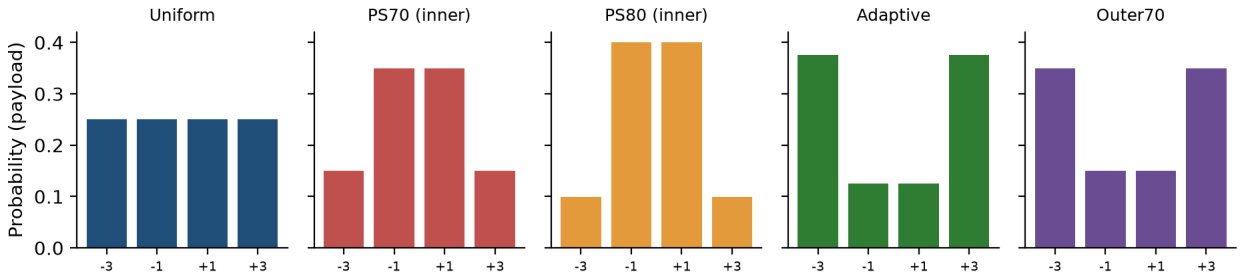
- Research collaboration between researchers at Manchester and Southampton
- Uniform, Fixed and Adaptive Shaped PAM4 for IM/DD: technical discussion deck
- SIMULATION RESULTS ONLY — NO SOUTHAMPTON MEASUREMENTS HAVE BEEN IMPORTED.

Problem statement

- Does shaping improve an unamplified IM/DD link, and under which constraint?
- Baud rate, entropy, net rate and optical power are separate quantities
- A fair comparison must control them explicitly
- Lower BER at equal baud is not more throughput

PAM4 fundamentals

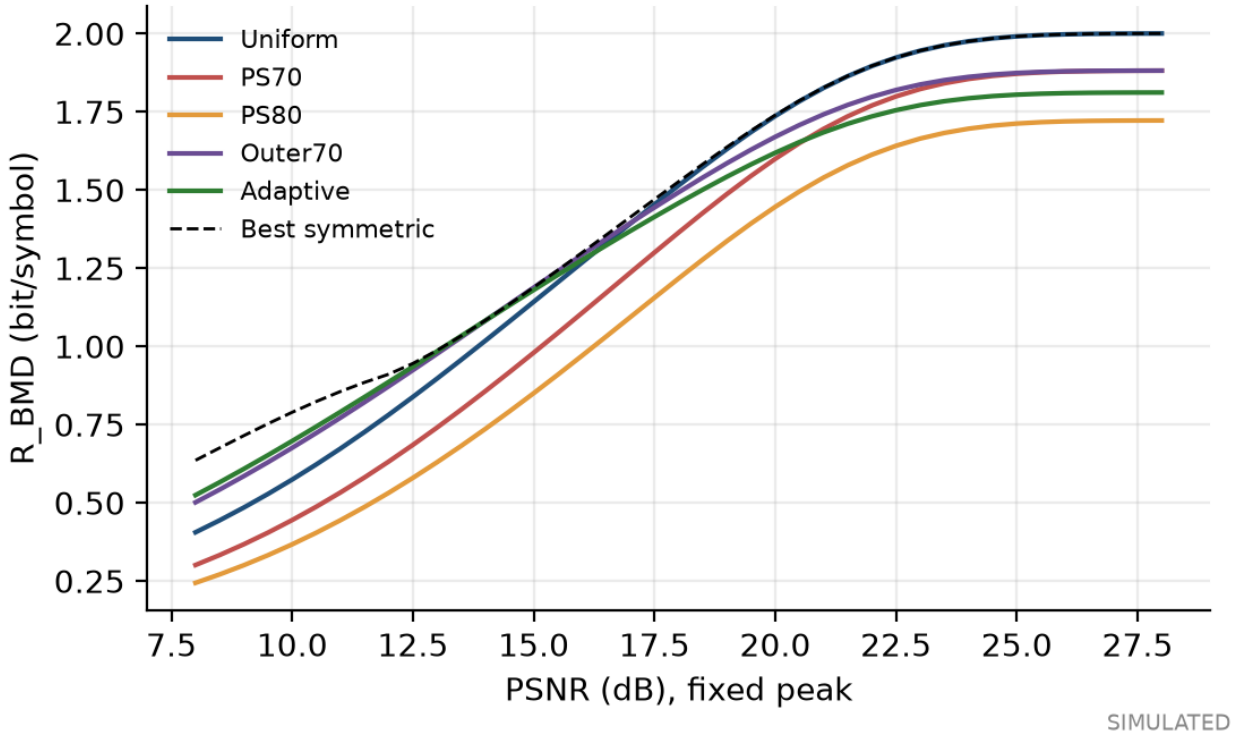
- Levels -3, -1, +1, +3 with Gray labels 00, 01, 11, 10
- Uniform carries 2 bit/symbol
- Fixed drive swing fixes level spacing, so noise margin stays the same
- Electrical energy is not optical power



SIMULATION ONLY: no Southampton measurements imported

Probabilistic shaping and the peak constraint

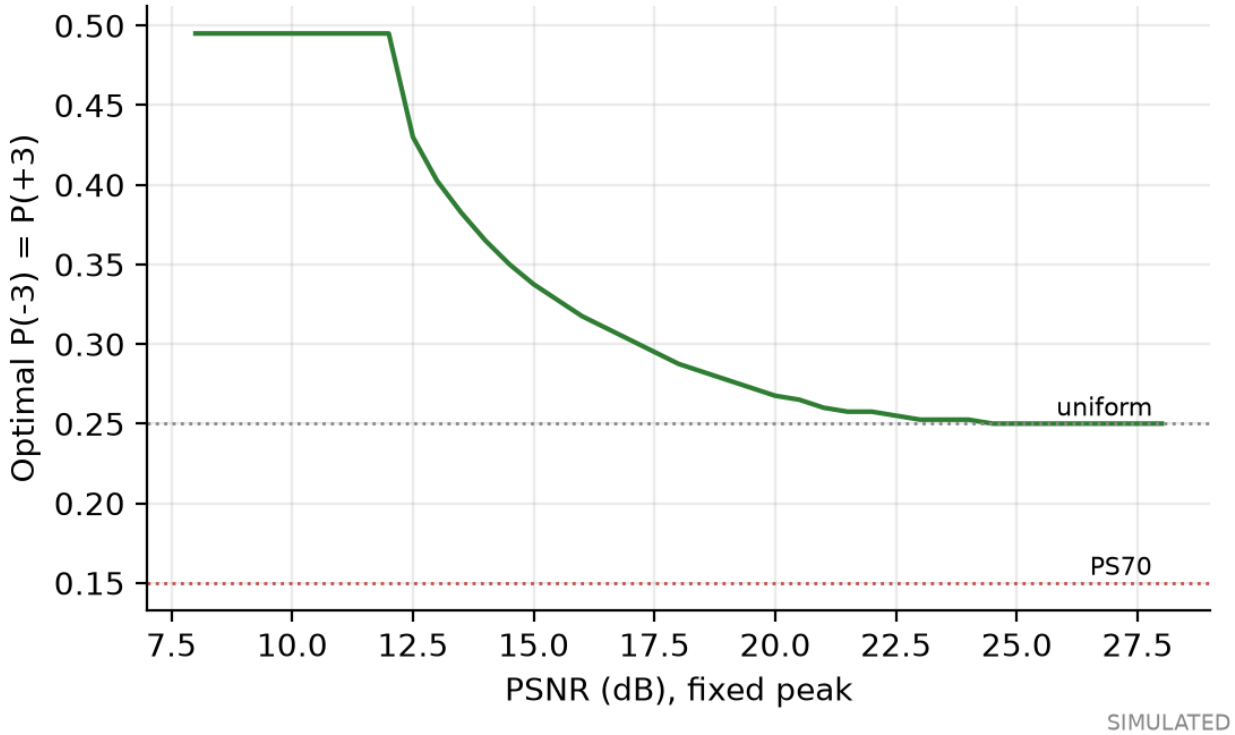
- Under an average-power limit, inner-heavy input helps
- A modulator's extinction ratio imposes a peak limit
- Then optimal input is outer-heavy (Wiegart et al., 2021)
- 14 dB, SMD: Uniform 1.035, PS70 0.869, PS80 0.745, Outer70 1.101



SIMULATION ONLY: no Southampton measurements imported

Maxwell-Boltzmann family

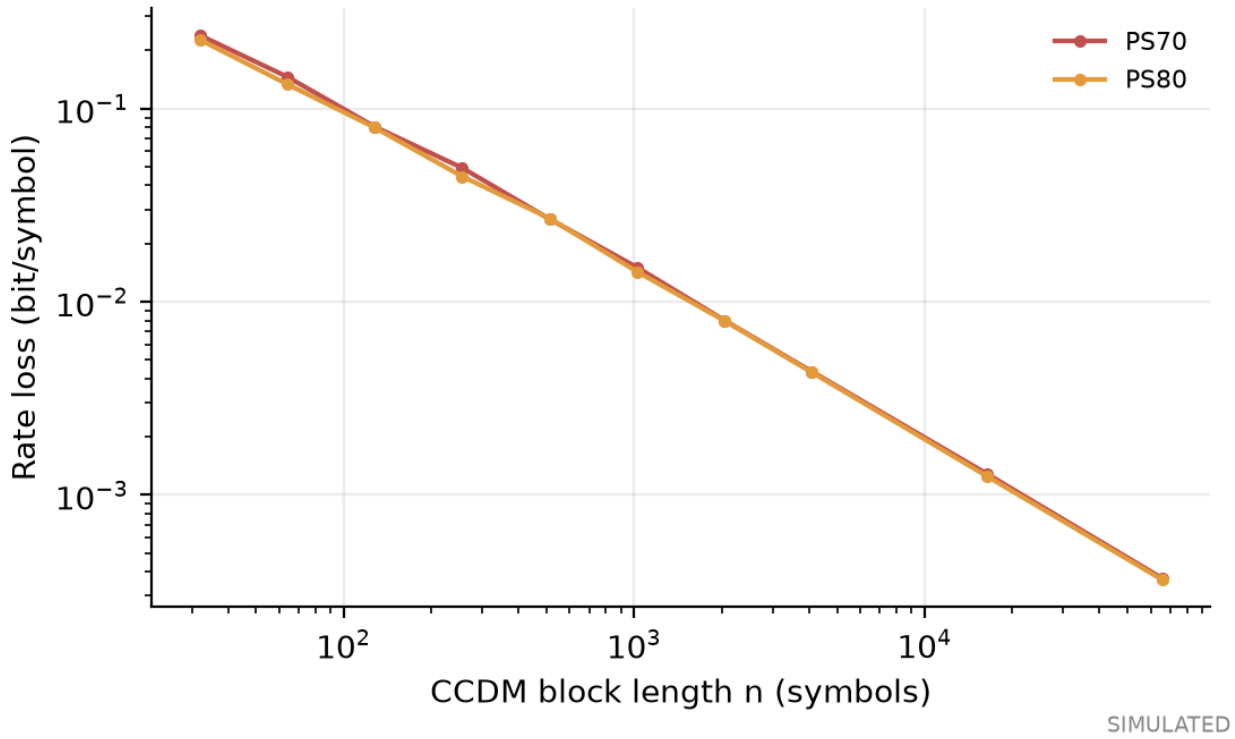
- $P(x) = \exp(-\lambda x^2) / Z$
- $\lambda = 0$ uniform; $\lambda > 0$ inner-heavy; $\lambda < 0$ outer-heavy
- PS70 and PS80 have $\lambda > 0$
- Optimum moves to uniform as PSNR rises



SIMULATION ONLY: no Southampton measurements imported

CCDM architecture

- Exact enumerative implementation, 1024-symbol blocks
- Rate loss: 0.015 bit/symbol for PS70
- One symbol error destroys a block
- No FEC, so this is not PAS



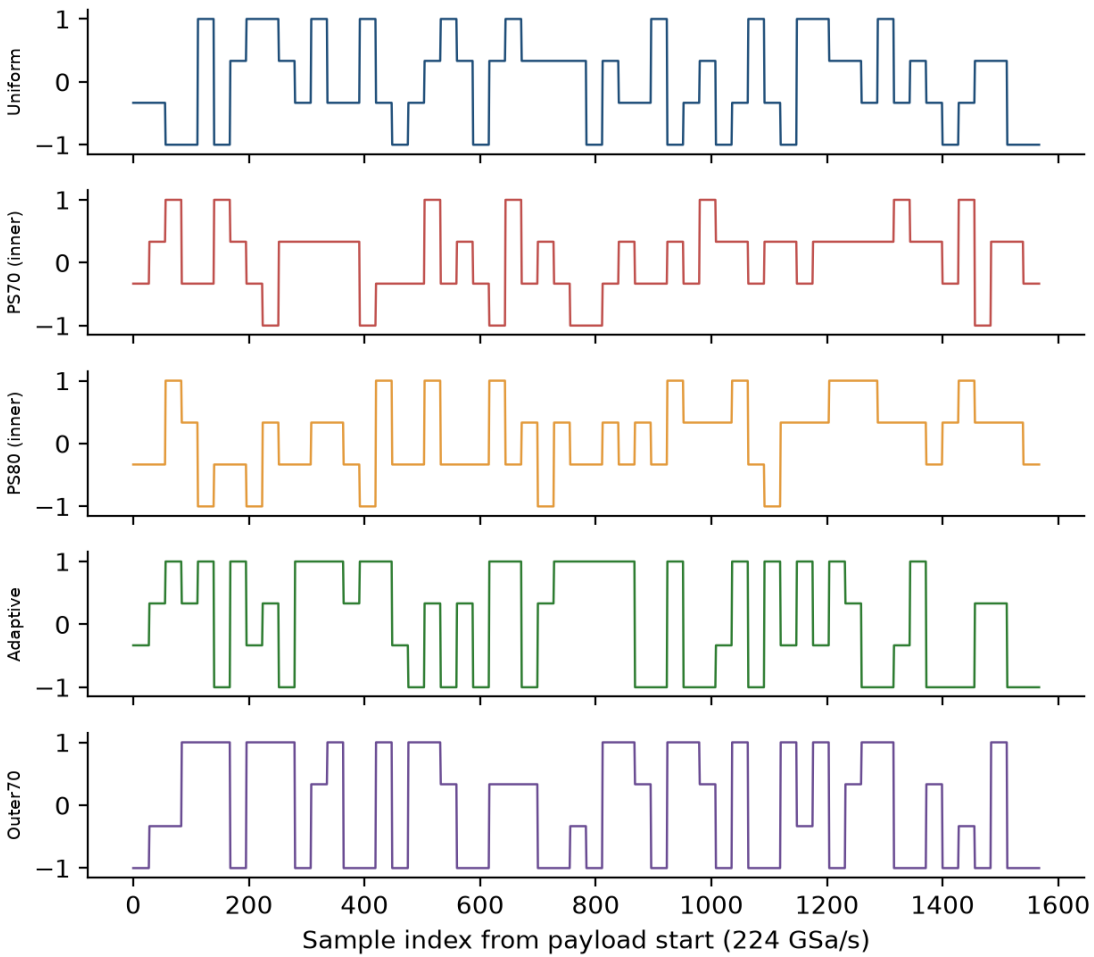
SIMULATION ONLY: no Southampton measurements imported

System architecture

- Bits → CCDM → Gray levels → frame → NRZ or RRC → M8199B CSV
- Link simulation or a real capture
- Sync → calibrate → FFE → MD or MAP → metrics
- Fitting uses preamble and training only

Keysight M8199B waveform package

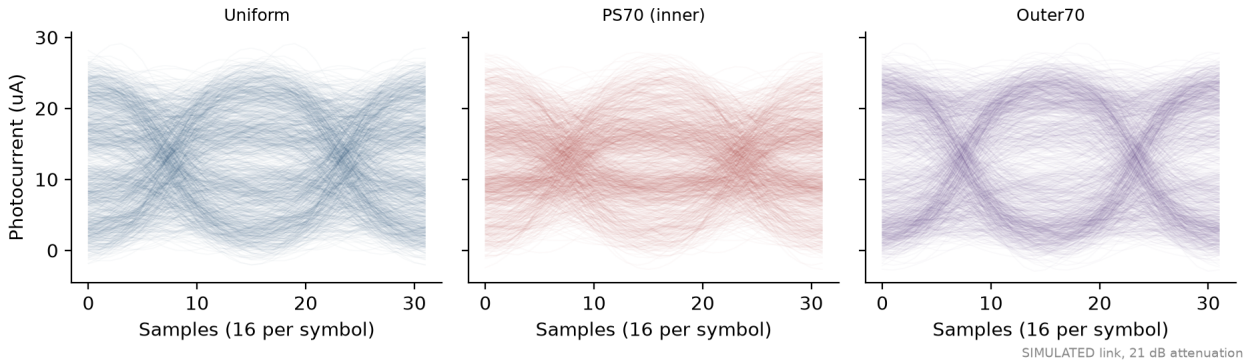
- Profile A: 224 GSa/s, 8 GBaud, 28 sps, 917,504 samples
- Profile B: 224 GSa/s, 56 GBaud, 4 sps, 262,144 samples
- Header DataOut1, within ±1, multiples of 512
- CSV rules from owner's notes; playback unverified



SIMULATION ONLY: no Southampton measurements imported

Receiver DSP

- Software sync with polarity and phase search
- 15-tap FFE trained on known symbols
- MAP: argmax p(y|x) P(x), with likelihoods from training
- Metrics: SER, label BER, R_SMD, R_BMD



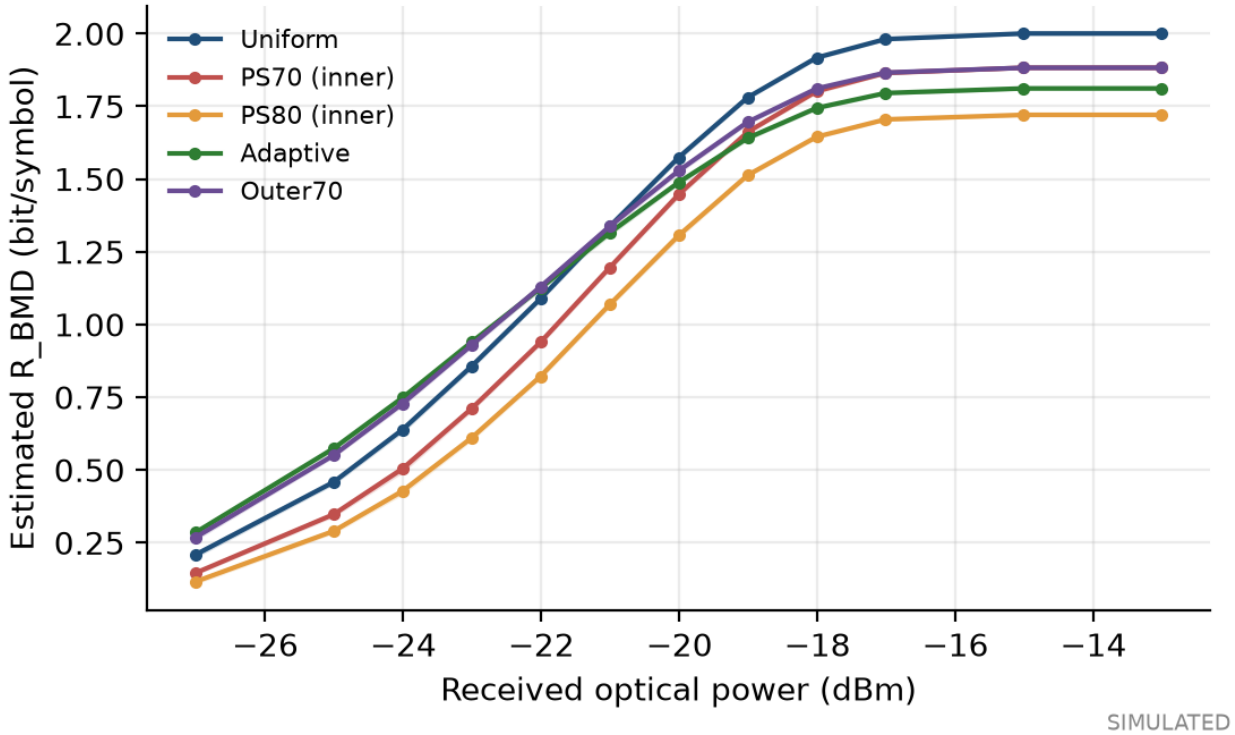
SIMULATION ONLY: no Southampton measurements imported

Experimental fairness

- Protocol A: equal baud, equal drive swing
- Protocol B: equal net rate, with any baud change reported
- Vpp, RMS, launch, received power, OMA, ER recorded separately
- Paired runs; bootstrap over runs, not samples

Baseline comparison, Profile A near -22.0 dBm

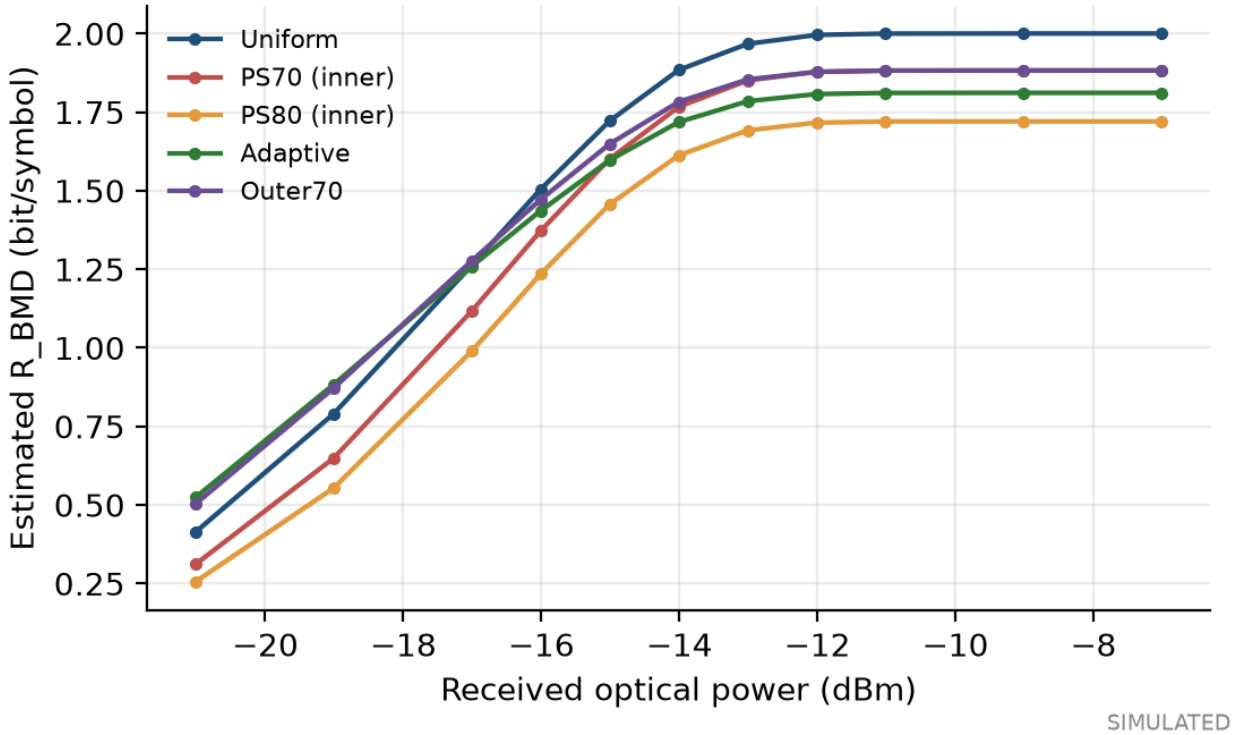
- PS70 (inner) vs Uniform: lower: -0.150 b/sym (95% CI -0.151 to -0.147)
- PS80 (inner) vs Uniform: lower: -0.267 b/sym (95% CI -0.270 to -0.264)
- Adaptive vs Uniform: higher by +0.036 b/sym (95% CI +0.033 to +0.039)
- Outer70 vs Uniform: higher by +0.041 b/sym (95% CI +0.037 to +0.045)



SIMULATION ONLY: no Southampton measurements imported

Profile B near -19.0 dBm

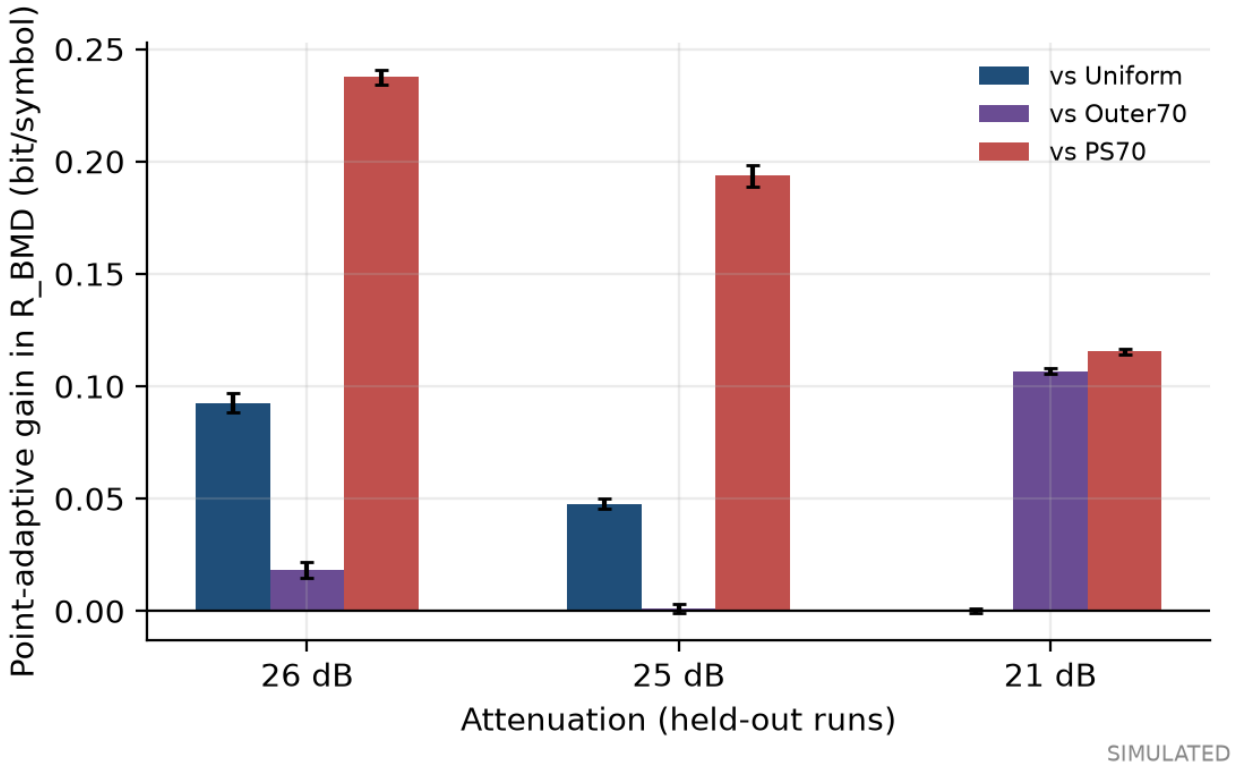
- PS70 (inner) vs Uniform: lower: -0.141 b/sym (95% CI -0.143 to -0.139)
- PS80 (inner) vs Uniform: lower: -0.236 b/sym (95% CI -0.238 to -0.234)
- Adaptive vs Uniform: higher by +0.093 b/sym (95% CI +0.090 to +0.097)
- Outer70 vs Uniform: higher by +0.081 b/sym (95% CI +0.078 to +0.084)



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Adaptive shaping on held-out runs

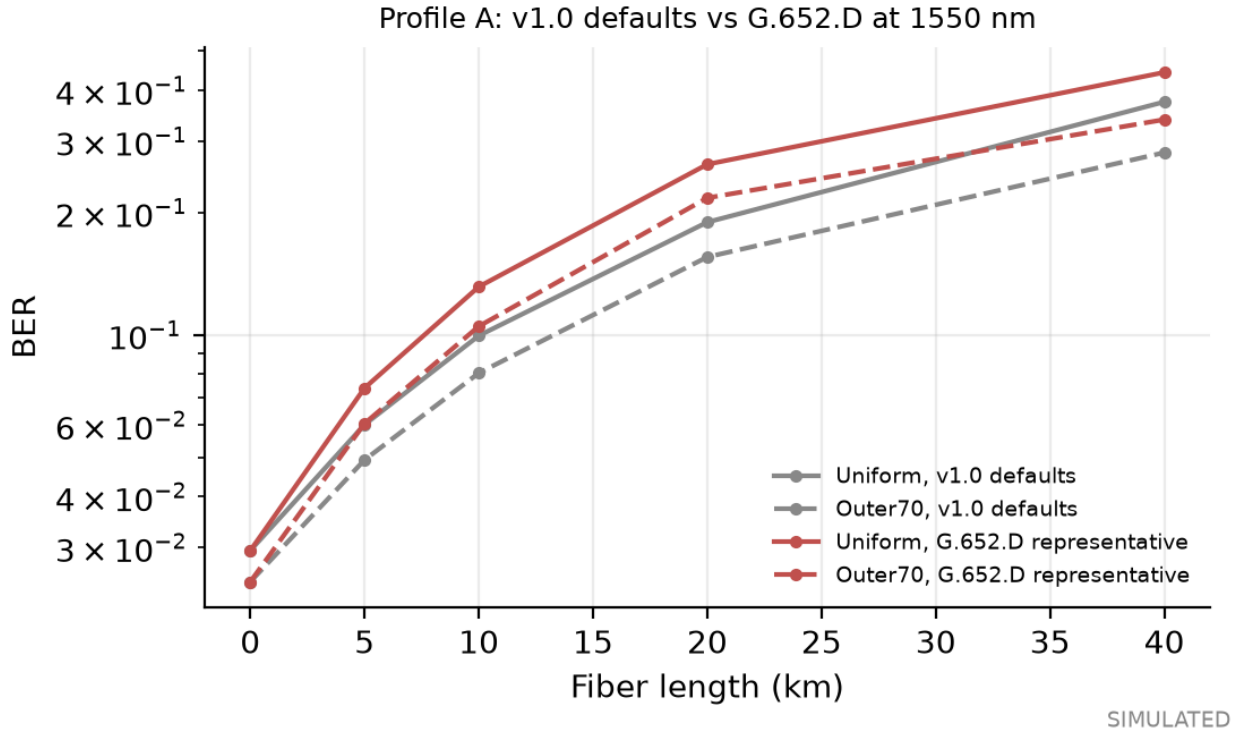
- Exported file: P(±3) = 0.375, simulation-optimized
- 26 dB: chose P(±3) = 0.43; vs Outer70 higher by +0.018 b/sym (95% CI +0.015 to +0.022)
- 25 dB: chose P(±3) = 0.34; vs Outer70 not resolved: +0.001 b/sym (95% CI -0.001 to +0.003)
- 21 dB: chose P(±3) = 0.25; vs Outer70 higher by +0.107 b/sym (95% CI +0.105 to +0.108)



SIMULATION ONLY: no Southampton measurements imported

Fiber model: ITU-T G.652.D presets

- Selectable fiber: none (v1.0), G.652.D representative, bounds low and high, custom
- Values carry provenance: limit, typical link value, data sheet or measured
- Re-run of the v1.0 fiber sweep: 25 of 25 Profile A points identical
- Dispersion null: simulation within 3e-08 of theory
- Attenuation 0.275 vs 0.2 dB/km drives the change; slope is negligible
- Analytical and simulated only; Southampton fiber type and length unconfirmed



SIMULATION ONLY: no Southampton measurements imported

Key findings and limitations

- PS70 and PS80 are expected to lose at fixed swing
- Outer-heavy is the candidate to measure
- No FEC: rates are achievable bounds
- Model parameters are generic; compression and AC effects need measurement

SIMULATION ONLY: no Southampton measurements imported

Southampton next steps

- Confirm receiver model and modulator limits
- Verify CSV import on the M8199B
- First run: Profile A, Uniform and Outer70, back-to-back
- Separate training and held-out captures, 5+ repeats
- Then regenerate this analysis from measurements